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Sathwik HS

Research Intern - PESU C-ISFCR

ABOUT

I'm a third year(2025-2026) student who likes building stuff that works well and feels right — mostly with React and JavaScript on the frontend, and sometimes with Solidity when it comes to secure, decentralized systems. I kinda sit at the intersection of UI design, blockchain tech, and computer networks — and I enjoy connecting the dots between them.

Lately, I've been working on a decentralized certification platform — something to fix how academic credentials are shared and verified. It should be simpler, safer, and something that actually makes sense in 2025, not stuck in the past. On the side, I've been messing around with cross-protocol bridging — figuring out how different networking protocols talk to each other (or don't).

I've got some certifications in Ethical Hacking and Frontend Development, but mostly I just like learning and building things that blend good UX with trust and transparency. If it involves users, networks, and making things a little more secure and smooth — I'm probably into it.

SKILLS

Technical Skills

Solidity, JavaScript, Web Development, C, Python, React.js, Computer Networks

Soft-Skills

Collaboration, Problem-solving, Communication, Time management, Result-oriented

Languages

English, Kannada, Hindi

CERTIFICATIONS

- Fullstack Web Development
- Ethical Hacking

- AI Azure Internship by Microsoft and Edunet

EXPERIENCE

PESU C-ISFCR, Bengaluru – *Blockchain Research Intern*

Jun 2025 – PRESENT

Contributing to a blockchain-based platform for issuing and verifying soulbound NFT certificates through secure NFT contracts, APIs, and React UI. Implemented issuer verification, Merkle tree issuance and QR verification for ensuring robust security.

- These are the things that is planned on this project :
- Deploy soulbound NFT contracts with Merkle tree verification, supporting a huge amount of users/issuers.
- Creating DNS/social media issuer verification, minting soulbound badges.
- Developing RESTful APIs with Pinata/IPFS for secure metadata storage.
- Building responsive React UI with custom CSS for role-based dashboards.
- Enabling QR-based certificate verification for instant on-chain validation.
- Enforced MetaMask authentication and admin access controls for security.
- Also, adding ZK placeholders and revocation metadata for future extensibility.

EDUCATION

PES University – *B.Tech*

Oct 2023 – Present, Bengaluru

TLC PU College (A Unit of CFAL) – *Pre-University Course*

May 2021 – March 2023, Mangaluru

PROJECTS

Bank Network – *Computer Networks*

Jan 2025 – Feb 2025

Bank Network – A simulated banking network built using Cisco Packet Tracer, featuring VLAN segmentation, static routing, firewall configurations. Designed to securely connect multiple bank branches with optimized performance, reliable remote access.

- Designed and simulated a comprehensive bank network using Cisco
- Packet Tracer, incorporating essential networking components and
- configurations.
- Configured VLANs and subnets to optimize network performance and

- ensure efficient data flow between different bank departments.
- Documented the network design and configurations in detailed
- reports, facilitating clear communication of the project's architecture
- and functionalities.

Avora – *Web Technology*

Aug 2024 – Nov 2024

Avora – A responsive web-based art marketplace developed using HTML, CSS, React, JavaScript. It allows artists to showcase, auction, and sell artworks, featuring seamless navigation, interactive bidding, user-friendly cart & wish list functionality.

- Developed a responsive web UI using HTML, CSS, React and JavaScript with intuitive navigation and cross-device compatibility.
- Implemented core features like artwork listings, cart, wishlist, and auction-style bidding.
- Simulated real-world marketplace functionality to enhance user engagement.

RESONATIA – *Python*

Nov 2024 – Dec 2024

Contributing to a blockchain-based platform for issuing and verifying soulbound NFT certificates through secure NFT contracts, APIs, and React UI. Implemented issuer verification, Merkle tree issuance and QR verification for ensuring robust security.

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